Seungjun (Jun) Cha

PhD student at Georgia Tech | AI for Materials
School of Materials Science and Engineering
School of Computational Science and Engineering
Atlanta, GA.

Email: scha@gatech.edu
Mobile: +1 (404) 706-6985
Google Scholar
LinkedIn: seungjun-cha
GitHub: sj-cha

I develop machine-learned interatomic potentials (MLIPs) and agentic framework for materials discovery. My work combines atomistic simulations and autonomous workflow to investigate materials surfaces/interfaces.

Education

- **Georgia Institute of Technology** — Atlanta, GA, USA
  Ph.D. in Computational Science and Engineering — Aug 2024 – May 2029 (Expected)
  - Advisors: Guoxiang (Emma) Hu, Victor Fung
- **Nanyang Technological University (NTU)** — Singapore
  B.Eng in Materials Engineering | First Class Honors — May 2024
  - Advisor: Kedar Hippalgaonkar

Skills

- **ML & Agents** – PyTorch, Pydantic-AI, MLIP (MatterSim, UMA, MACE, etc.) – fine-tuning, architecture development, Multi-agent orchestration, ReAct loop, Bayesian optimization
- **Scientific Computing** – VASP, ASE, Pymatgen, RDKit, NVIDIA Alchemi, HPC (Slurm, GPU computing)
- **Programming** – Python, C, Bash, Linux, Git
- **Languages** – English (Native), Korean (Native)

Research Experience

- **Graduate Trainee** — Seattle, WA, USA
  NSF Center for Integration of Modern Optoelectronic Materials on Demand (IMOD) — Oct 2025 – Present
  - Designed PerovAgent, a multi-agent framework combining LLM-generated ligand hypotheses, dispersion-corrected MLIP screening, and matched-pair statistical analysis; corroborated a new ligand-design principle with DFT.
  - Developed PerovKit, a Python library that automates construction of ligand-passivated nanocrystal models.
  - Investigated how zwitterionic ligand headgroup geometry influences binding-mode fluctuations using MLIP-based molecular dynamics, suggesting a possible mechanism for photoluminescence blinking.
  - Revealed that spacer-spacer interaction can drive thermodynamic instability and phase segregation in spacer-mixed 2D perovskites using DFT. (Under review, *Adv. Mat.*)
- **Graduate Research Assistant** — Atlanta, GA, USA
  School of Materials Science and Engineering, Georgia Institute of Technology — Aug 2024 – Present
  - Developing a recursive self-improvement (RSI) pipeline in which an LLM learns physically valid hybrid 2D perovskite structures from a calibrated, physics-based verifier reward alone, without MLIP or DFT.
  - Built Paper2Structure, a multi-agent pipeline that analyzes texts/figures from an experimental paper, writes a build script for an atomistic model via a sandboxed coding agent, and validates the model against the paper's claims.
  - Devised and implemented GC-QEq, a grand canonical charge equilibration architecture for MLIPs, conditioned on electrode potential; demonstrated the simulation of voltage-dependent IR spectra of solvated MoS_2.
  - Fine-tuned a universal MLIP for realistic ligand-passivated $CsPbBr_3$ nanocrystals, revealing ligand and site-dependent phonon dynamics and nonmonotonic lattice stiffening with ligand binding strength (*JACS Au* 2026).
  - Established a nonmonotonic relationship between ligand binding energy and photoluminescence quantum yield, deriving experimentally validated design principles for $CsPbX_3$ nanocrystals (*Chem. Mater.* 2026).
- **Materials Engineer Intern** — Singapore
  Cambridge Center for Advanced Research and Education in Singapore (CARES) — Jan 2023 – May 2023
  - Conducted synthesis and characterization of ZnO nanoparticles and Bayesian optimization of synthesis parameters for improved UV absorption.

- **Research Intern** Seoul, South Korea
  Department of Materials Science and Engineering, Seoul National University Jun 2022 – Aug 2022
  - Modeled crystal plasticity of aluminum alloys and determined microstructure constants via Bayesian optimization.

Publications (Google Scholar)

Published

- Cha, S; Wang, C*; Fung, V*; Hu, G*. Ligand-Controlled Phonon Dynamics in $CsPbBr_3$ Nanocrystals Revealed by Machine-Learned Interatomic Potentials. *JACS Au* 2026, 6 (5), 3039–3047.
- Cha, S#; Brea, C#; Malinoski, A#; Wang, C*; Hu, G*. Design Principles for Surface-Passivating Ligands of Cesium Lead Halide Perovskite Nanocrystals in the Strongly Quantum-Confined Regime. *Chem. Mater.* 2026, 38(7), 3429–3438.
- LaFollette, DK; Gomez-Dominguez, M; Datta, K; Conde del Moral, A; Cha, S; Lawton, J; Li, R; Lawrie, B; Luo, Y; Perini, CAR; Hu, G; Correa-Baena, JP*. Suppressing Phase Segregation and Improving Stability in Mixed-Halide Perovskites through Spinel Oxide-Directed Epitaxy. *J. Am. Chem. Soc.* 2026, 148 (25), 26076–26091.
- Nyiera, HN; Yang, X; Liu, Y; Cha, S; Chen, O; Hu, G; Chen, K; Zhao, J*. Correlating Intensity Fluctuations and Lifetime Distributions Reveals Surface Chemistry Effects on Photoluminescence of Single $CsPbBr_3$ Quantum Dots. *ACS Appl. Mater. Interfaces* 2026, 18 (23), 32997–33006.

Under review/In preparation

- Cha, S; Jeong, YJ; Conde del Moral, A; Correa-Baena, JP; Marder, S; Ginger, D*; Fung, V*; Hu, G*. Autonomous Discovery of Ligand Design Principles for Halide Perovskite Surfaces. *Nat. Commun.*
- LaFollette, DK; Gomez-Dominguez, M; Cha, S; Conde del Moral, A; Mohamed, A; Lawton, J; Perini, CAR; Kandada, ARS; Hu, G; Correa-Baena, JP*. Organic Cation Chemistry Drives Phase Segregation, Rigidity, Dimensionality, and Stability in 2D and Quasi-2D Perovskites. *Adv. Mater.*
- Gomez-Dominguez, M; Kumar, E; Watkins, N; Koch, K; Tasci, C; Cha, S; Hu, G; Flick, J; Menon, V; Kandada, ARS; Correa-Baena, JP*. Engineering Exciton-Polariton Dispersion Through Structural Handles. *ACS Nano*
- Lee, W; Kim, S; Fitzgerald, M; Cha, S; Shallenberger, J; Hu, G; Doumon, N*; Asuo, I*. Gas Quenching-Mediated Crystallization of Compositionally and Interfacially Engineered Semitransparent Perovskite Solar Cells with Over 6% Potential Light Utilization Efficiency. *Carbon Energy*
- Bansal, A#; Cha, S#; Fung, V; Hu, G*. Constant-Potential Machine Learning Force Field for Electrochemical Interfaces.

Honors/Awards

- **MSE-External Advisory Board (EAB) Fellowship** Aug 2026
- **JACS Au Early Career Advisory Board (ECAB) Selects** Jun 2026
  - First-author paper selected for the *JACS Au* ECAB Selects Special Collection.
- **Best Poster Award**, 2025 GT MSE Fall Poster Competition Oct 2025
  - 1st place in *Materials AI & Accelerated Materials Discovery* division
- **Dean's List**, School of Materials Science and Engineering, NTU May 2024
- **Global Hansang Dream Scholarship**, Global Hansang Dream Foundation Dec 2022
- **President Research Scholar**, NTU Aug 2020
- **MOE Tuition Grant**, Ministry of Education, Singapore Aug 2018

Presentations

- 12 oral/poster presentations, including two invited talks, at conferences such as MRS, ACS, and SMLQC.

Teaching / Service

- **President**, Materials Research Society Georgia Tech Chapter Apr 2026 – Present
- **Financial Officer**, Materials Research Society Georgia Tech Chapter Apr 2025 – Mar 2026
- **Symposium Assistant (MT01)**, MRS Spring Meeting Apr 2025
- **Founder/Planning Executive**, NTU Korean Student Association Aug 2019 – Jun 2020